Document Information:

Field	Details
Document Name	Problem Statement
Project Name	Student Performance Analytics & Decision Support System
Client	Mega Central High School
Project Sponsor	Chairman, Mega Central High School
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1. Introduction:

Mega Central High School conducts periodic academic assessments for Grade 10 students across four core subjects:

- Mathematics
- Science
- English
- Social Science

Currently, assessment activities are conducted through traditional offline classroom methods. Teachers evaluate answer sheets manually and maintain student performance records using individual Excel spreadsheets.

While this approach supports basic academic record maintenance, the school lacks a centralized mechanism to analyze student performance, monitor academic progress, identify learning gaps, and support timely intervention.

2. Current Situation:

The existing academic reporting process involves:

1. Teachers conduct classroom assessments.
2. Students complete written examinations.
3. Teachers manually evaluate answer sheets.
4. Marks are recorded in individual spreadsheets.
5. Subject-wise reports are prepared separately.
6. Results are consolidated manually.
7. Management reviews reports periodically.

Each department maintains its own records, resulting in fragmented academic information across multiple files.

3. Problem Statement:

Mega Central High School currently does not have a centralized Student Performance Analytics system to consolidate and analyze academic assessment information.

The existing spreadsheet-based approach creates several challenges:

- Student performance data is distributed across multiple files.
- Teachers spend significant time preparing academic reports.
- Manual calculations increase the possibility of errors.
- School management lacks timely visibility into student performance.
- Students requiring academic support are identified only after major examinations.
- Chapter-level learning gaps are difficult to identify.
- Historical performance trends cannot be effectively analyzed.

As a result, academic decisions are often reactive rather than proactive.

4. Business Impact:

Existing Problem	Business Impact
Multiple assessment spreadsheets	Data inconsistency and duplication
Manual score calculation	Increased teacher workload
Manual report preparation	Delayed academic reporting
No centralized analytics system	Limited management visibility
Late identification of weak students	Delayed academic intervention
Lack of chapter analysis	Difficulty improving teaching strategies
No performance trends	Limited understanding of student progress

5. Root Cause Summary:

People Related Causes

- Teachers maintain independent spreadsheets.
- Reporting formats differ between departments.
- Academic analysis depends heavily on manual effort.

Process Related Causes

- No standardized assessment reporting workflow.
- Performance monitoring happens periodically rather than continuously.
- Reports are mainly generated after examinations.

Technology Related Causes

- Dependency on Excel-based reporting.

- No centralized academic database.
- No automated analytics and visualization platform.

Data Related Causes

- Assessment data exists in multiple locations.
- Lack of standardized data structure.
- Limited data validation mechanisms.
- Difficulty maintaining historical records.

6. Business Need:

The school requires a centralized analytics solution that enables:

- Consolidation of assessment data.
- Automated score and grade calculation.
- Continuous student performance monitoring.
- Early identification of academically at-risk students.
- Subject and chapter-level performance analysis.
- Data-driven academic decision-making.

7. Proposed Solution:

The Student Performance Analytics & Decision Support System will:

- Create a centralized academic data repository.
- Automate assessment score calculations.
- Track student performance across multiple tests.
- Analyze question-level and chapter-level outcomes.
- Identify students requiring intervention.
- Provide interactive dashboards using Microsoft Power BI.
- Support teachers and management with actionable insights.

8. Expected Business Outcomes:

Outcome	Expected Improvement
Academic reporting	Faster and standardized
Score calculation	Automated
Student monitoring	Continuous
Risk identification	Earlier detection
Decision making	Data-driven
Teacher workload	Reduced

9. Success Measurement:

The project will be considered successful when:

KPI	Target
Dashboard availability	100%
Manual score calculation	Eliminated
Report preparation effort	Reduced by 80%
High-risk student identification	Before final exams
Teacher dashboard adoption	Greater than 90%

10. Conclusion:

The absence of a centralized academic analytics platform limits Mega Central High School's ability to proactively monitor student performance and improve learning outcomes.

The proposed Student Performance Analytics & Decision Support System will transform manually maintained assessment data into meaningful insights, enabling teachers and management to make timely, evidence-based academic decisions.